Sensor Configuration File

Team Equinox — aBAJA SAEINDIA 2026 | Autonomous Vehicle System

1. Overview

This document provides the complete sensor configuration, embedded hardware integration, vehicle parameter configuration, and autonomous perception architecture used in the Team Equinox autonomous vehicle platform for aBAJA SAEINDIA 2026. The autonomous stack is designed for real-world deployability using ROS 2 Humble middleware, embedded NVIDIA Jetson compute hardware, and automotive-compatible sensing systems. The architecture integrates camera-based lane perception, radar-based obstacle detection, vehicle state estimation sensors, and Extended Kalman Filter (EKF) based sensor fusion. The system has been developed to satisfy real-time autonomous operation requirements while maintaining scalability for future deep learning and advanced autonomous navigation upgrades.

2. Camera Perception Sensor

The primary perception subsystem for Lane Keeping Assist (LKA) utilizes a forward-facing RGB camera. The camera stream is processed using OpenCV-based computer vision algorithms operating on ROS 2 Humble nodes. The perception pipeline includes: • Image acquisition • Grayscale conversion • Binary thresholding • Perspective transformation (Bird's-Eye View) • Sliding window lane search • Polynomial lane fitting • Lane center estimation • PID steering correction generation The camera subsystem operates at approximately 30 FPS and is optimized for embedded deployment on the NVIDIA Jetson Orin Nano 8GB platform.

Parameter	Value	Description
Sensor Type	RGB Camera	Primary perception sensor
Resolution	640 × 480	Real-time embedded processing
Frame Rate	~30 FPS	Synchronized with control loop
Mount Position	Front Forward Facing	Lane perception and navigation
Processing Framework	OpenCV + ROS 2	Computer vision pipeline
Primary Function	Lane Detection	LKA system perception

3. Vehicle State Sensors

The autonomous platform integrates multiple vehicle state estimation sensors for reliable motion estimation and autonomous navigation. The vehicle utilizes: • IMU for orientation and angular velocity estimation • Wheel speed sensing using Hall Effect sensors • Steering position sensing using an Absolute Encoder • Wheel encoder systems for odometry estimation • Hall Effect sensors integrated into the motor controller for rotational feedback These measurements are fused using an Extended Kalman Filter (EKF) architecture to improve localization accuracy, noise rejection, and dynamic vehicle state estimation.

Sensor	Configuration	Purpose

IMU	Inertial Measurement Unit	Orientation and motion estimation
Wheel Speed Sensor	Hall Effect Sensor	Wheel rotational speed estimation
Steering Encoder	Absolute Encoder	Steering angle feedback
Wheel Encoder	Rotary Encoder	Odometry and velocity estimation
Motor Controller Sensor	Hall Effect Sensor	Motor rotational feedback
Sensor Fusion	Extended Kalman Filter (EKF)	State estimation and localization

4. Radar Obstacle Detection

The Autonomous Emergency Braking (AEB) subsystem integrates automotive radar sensing for obstacle detection and collision avoidance. The radar subsystem provides: • Obstacle range estimation • Relative velocity estimation • Collision probability analysis • Low visibility operational robustness • Dynamic obstacle tracking Radar data is processed through a fuzzy logic-based controller to determine braking intensity and emergency intervention conditions.

Parameter	Value	Purpose
Sensor Type	Automotive Radar	Obstacle detection
Detection Function	Distance + Velocity	Collision analysis
Control Algorithm	Fuzzy Logic	Dynamic braking decision
Primary Application	AEB	Emergency braking

5. Embedded Compute Platform

The autonomous system is deployed on an embedded robotics platform based on the NVIDIA Jetson Orin Nano 8GB. The software architecture integrates: • Ubuntu Server (Canonical) • ROS 2 Humble • OpenCV perception pipelines • EKF sensor fusion framework • PID lane control • Fuzzy logic braking control • RViz visualization tools • IPG CarMaker simulation • CARLA simulation environment

Parameter	Value	Use
Compute Platform	NVIDIA Jetson Orin Nano 8GB	Embedded autonomous computation
Operating System	Ubuntu Server	ROS 2 deployment platform
Middleware	ROS 2 Humble	Distributed robotics communication
Power Supply	19V Regulated Supply	Vehicle compatible power input
Simulation Platforms	IPG CarMaker, CARLA, RViz	Testing and validation
Control Frequency	50 Hz	Real-time autonomous control
Perception Frequency	~30 FPS	Camera processing pipeline
EKF Frequency	100 Hz	Sensor fusion updates

6. Compliance with Autonomous Event Rules

The Team Equinox autonomous platform complies with the autonomous event requirements specified by aBAJA SAEINDIA 2026. The architecture ensures: • Real-world deployability • Approved sensor categories • Embedded compute compatibility • Real-time autonomous operation • ROS 2 modular architecture • Safe autonomous control mechanisms

Rule Requirement	Status
Approved autonomous sensor categories	✔ COMPLIANT
ROS 2 modular software architecture	✔ COMPLIANT
Embedded autonomous compute platform	✔ COMPLIANT
Real-time autonomous operation	✔ COMPLIANT
Sensor fusion architecture	✔ COMPLIANT
Safe autonomous control pipeline	✔ COMPLIANT